

SOFTWARE ENGINEERING – LEVEL 3 MINI-PROJECT (UE24CS341A)
SYNOPSIS / PROJECT PROPOSAL

1. Project Title and Group

Field	Details
Project Title	AttendEase – Student Attendance Management System
Group Number	Team 3
Course	UE24CS341A – Software Engineering
Semester	V Semester
Development Methodology	Agile (Scrum, 6 sprints)
Version	1.0
Submission Format	PDF

2. Team Members and SRNs

Sl.	Student Name	USN	SRN / PRN
1	Anshdeep Singh Sachdeva	PES1UG24CS070	PES1202400437
2	Arundhati K	PES1UG24CS083	PES1202402135
3	Atharva Ashish Vyas	PES1UG24CS093	PES1202400342
4	Bhoomika Dayanand Jituri	PES1UG25CS807	PES1202502970

Note: Every team member is involved in all project activities (requirements, design, implementation, testing, and documentation), in addition to owning one complete functionality as detailed in Section 8.

3. Problem Statement

Attendance in most college classrooms is still tracked manually — through roll calls or paper/spreadsheet registers maintained by individual faculty members. This approach is time-consuming during lecture hours, prone to transcription errors, difficult to consolidate across subjects and sections, and gives students no real-time visibility into where they stand. Attendance shortages are typically discovered only close to examinations, by which time students have little opportunity to recover, and the college office has no simple way to generate accurate, defensible, subject-wise defaulter lists.

4. Proposed Project Description

AttendEase is a web-based Student Attendance Management System that digitizes the end-to-end attendance workflow for a department. Administrators configure courses, sections and timetables and assign faculty to subjects. Faculty start an attendance session for each scheduled class and mark students present or absent in a few clicks. Students get a live, subject-wise and cumulative view of their attendance and can raise a correction request if a session was marked incorrectly, which the concerned faculty member can approve or reject. The system automatically computes attendance percentages after every session and flags students who fall below a configurable minimum threshold, and lets faculty/administrators export consolidated and defaulter reports.

This is a new, self-contained system proposed by the team (no existing customer engagement); the department's own attendance workflow is used as the reference scenario. At the conclusion of this project, a user of the system will be able to do the following (functional features):

- Register and log in with a role-appropriate account (Student, Faculty, or Administrator).

- As an administrator, create and manage courses, sections, timetables, and faculty-subject assignments.
- As a faculty member, start an attendance session for a scheduled class and mark each enrolled student present or absent.
- As a student, view subject-wise and cumulative attendance percentage and session-wise history.
- As a student, raise an attendance correction request; as a faculty member, approve or reject it.
- As a faculty member or administrator, generate and export a defaulter list and consolidated attendance report.
- As an administrator, view system-wide operational statistics and manage user accounts.

5. Target Users / Stakeholders

Stakeholder	Interest / Need
Students	Track their own attendance in real time and raise corrections when needed.
Faculty	Mark attendance quickly and reliably, review and act on correction requests.
Department Administrator / Class Coordinator	Configure courses/sections/timetables, monitor attendance compliance department-wide.

6. Objectives

- Reduce the time faculty spend marking attendance for a class session.
- Give students real-time, self-service visibility into their subject-wise attendance percentage.
- Maintain accurate, auditable attendance records with a controlled correction workflow.
- Enable administrators to configure courses, sections, timetables, and faculty-subject mapping.
- Automatically identify students below the minimum attendance threshold (default 75%) per subject.
- Develop the product incrementally using Agile practices, with CI/CD, code review and automated testing.

7. Scope

In Scope	Out of Scope
Student / Faculty / Admin registration and login	Biometric or RFID hardware-based attendance capture
Course, section and timetable management	Native mobile application
Faculty-subject-section assignment	Integration with a university ERP or examination system
Attendance session creation and marking	SMS / push notification gateway (email notifications only)
Attendance correction request and approval workflow	Multi-campus / multi-institution deployment
Student attendance dashboard (subject-wise and cumulative)	Parent-facing portal
Defaulter identification and report export (CSV/PDF)	Automated timetable clash resolution beyond simple overlap checks

8. Functional Features and Team-wise Ownership

Product ownership: each team member owns at least one complete functional feature end-to-end (requirements → design → implementation → testing). UI styling, database interaction and unit testing are supporting responsibilities, not substitutes for owning a functionality.

Member	USN	Owned Functional Feature(s)	Secondary Responsibility
Anshdeep Singh Sachdeva	PES1UG24CS070	F1. User Registration & Role-Based Authentication F3. Faculty–Course–Section Assignment	CI/CD pipeline & security validation

Member	USN	Owned Functional Feature(s)	Secondary Responsibility
Arundhati K	PES1UG24CS083	F2. Course & Timetable Management F4. Class Session Creation & Attendance Marking	UI/UX design
Atharva Ashish Vyas	PES1UG24CS093	F5. Student Attendance Dashboard F6. Attendance Correction Request & Approval	Database design
Bhoomika Dayanand Jituri	PES1UG25CS807	F7. Defaulter Identification & Report Generation F8. Admin Operational Dashboard & Data Export	Testing / QA & documentation

9. Plan of Work (Short-Term / Sprint Plan)

Sprint	Planned Output	Acceptance Evidence
Sprint 1	Project setup, requirements baseline, DB schema, auth module	User stories + Git repo + working login
Sprint 2	Course, section, timetable and faculty-assignment management	Demo + unit tests
Sprint 3	Attendance session creation and marking module	Integration tests + PRs
Sprint 4	Student dashboard and correction request/approval workflow	System demo + test evidence
Sprint 5	Defaulter reports, admin dashboard, export, security hardening	Validation report + coverage report
Sprint 6	System validation, performance/security testing, final release	Final report + deployment/demo

10. Proposed Technology / Tools

Category	Tool / Technology
Backend	Python 3.x, Django + Django REST Framework
Database	PostgreSQL (MySQL as fallback)
Frontend	Django templates with Bootstrap 5 (responsive UI)
Version Control	Git + GitHub
Agile / Project Management	GitHub Projects (backlog, sprints, issues)
CI/CD	GitHub Actions
Containerization	Docker
Testing	pytest / Django TestCase, coverage.py
Static Analysis	SonarQube (or flake8/pylint)
API Testing	Postman
Design / Diagramming	draw.io, PlantUML

11. Agile Justification

Agile/Scrum is used because requirements for an attendance system are best refined through short, demonstrable increments (auth → course setup → marking → dashboards → reports), each of which can be independently reviewed by the course instructor and validated against real usage scenarios. Weekly sprint reviews also make individual contribution and functional ownership easy to evaluate, as required by the project guidelines.